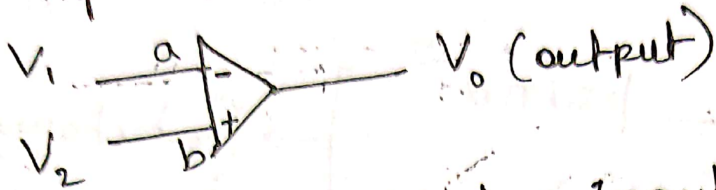


Operational Amplifiers (OP-Amp)

- Op-amps are building blocks of any integrated circuit.

2 inputs, 1 output.



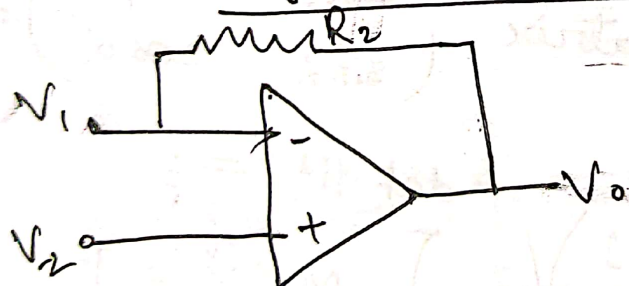
2 inputs, one output; Inputs $\rightarrow$ a: inverting input (-)

If we have a sinusoidal wave that comes through it, crest will look like a trough and trough like a crest,

b: non-inverting input,

- Ideal op-amp: Gain is infinite.

Negative Feedback:



$\Rightarrow$ output fed back to the inverting input.

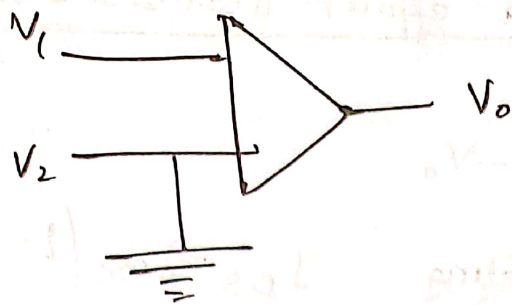
Differential amplifier:

Gain given by:

$$V_0 = A(V_2 - V_1)$$

gain (∞) $\rightarrow$ difference

For gain to be ∞ , $(V_2 - V_1)$ should be negligibly small.



Here, $V_2 = 0$

$$\Rightarrow V_1 = 0$$

▢ This is called a virtual ground.

[not directly grounding V_1 , but due to infinite gain, $V_2 - V_1 \rightarrow 0$, hence V_1 must be identically zero]

▢ When $V_2 = \frac{A}{A+1} V_1$ is finite, $\frac{V_2 - V_1}{A}$ also becomes finite non-zero.

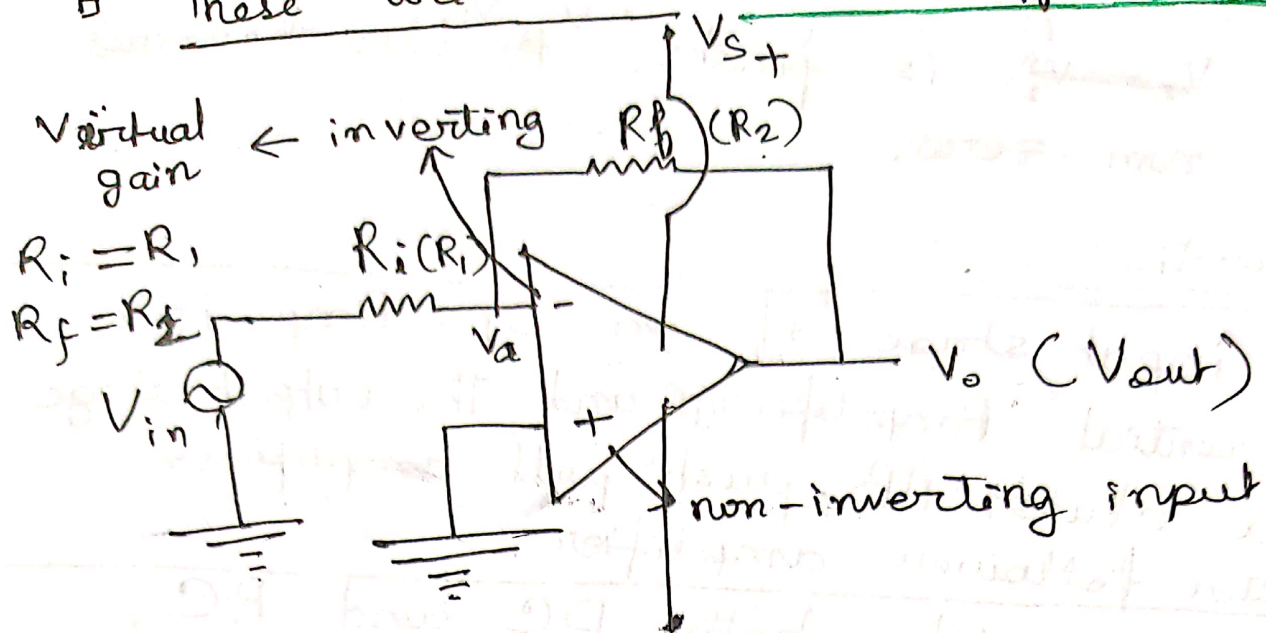
▢ Properties :-

- ▢ The input stage of an Op-Amp is a differential Amplifier (DA) and the output stage is a class AB push-pull ~~amplifier~~ emitter follower amplifier.
- ▢ It can amplify both DC and AC.
- ▢ Op-Amp has very high input resistance (ideally infinite) and very low output impedance.
- ▢ Op-Amp has a very high open-loop (no feedback) gain $\rightarrow$ ideally ∞ .

◦ High input resistance $\rightarrow 10^5 - 10^6 \text{ ohm}$

◦ High gain $\rightarrow 10^5 - 10^6$

□ These are all Field effect Transistor.



□ Case 1 (OP-Amp: $V_s \rightarrow \infty$)

Here, $V_o = A(V_a - V_b) = A V_{in}$

$V_b = 0$
 $V_a = V_{in}$

$I_{in} = \frac{V_{in}}{R_i}$ (i)

$\Rightarrow I_{in} = \frac{V_{in} - V_a}{R_i}$

$[V_a = 0]$

$\Rightarrow I_f = \frac{V_a - V_o}{R_f}$

$\Rightarrow I_f = -\frac{V_o}{R_f}$ (ii)

$A = \frac{V_o}{V_{in}} = -\frac{I_f R_f}{I_{in} R_i} = -\frac{R_f}{R_i}$

If the current flows entirely through the circuit and not through the op-amp.

$$I_i = I_f \quad [\text{as nothing goes through op-amp}]$$

Case II:

What if $A \neq \infty$, $A \rightarrow \text{finite}$

$$V_o = A(V_2 - V_1) \quad \text{if,} \quad V_1 = -\frac{V_o}{A} \quad \text{--- (i)}$$

(We ground V_2)

When no virtual grounding of V_1 and, and finite A , ($V_2 \neq V_1$)

Now,

$$I_{in} = I_f = \frac{V_{in} - V_1}{R_i} = \frac{V_{in} + \frac{V_o}{A}}{R_i}$$

$$I_{in} = I_f = \frac{V_{in} + \frac{V_o}{A}}{R_i} \quad \text{--- (ii)}$$

$$V_o = V_1 - I_f R_f = -\frac{V_o}{A} - \left(\frac{V_{in} + \left(\frac{V_o}{A} \right)}{R_i} \right) R_f$$

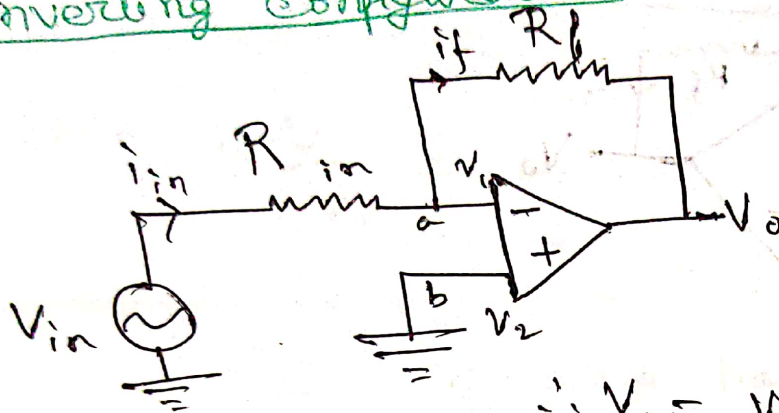
$$\Rightarrow V_o \left(1 + \frac{1}{A} + \frac{1}{A} \frac{R_f}{R_i} \right)$$

$$= -V_{in} \frac{R_f}{R_i}$$

$$\text{Gain} = \frac{V_o}{V_{in}} = \frac{\left(-\frac{R_f}{R_i} \right)}{\left(1 + \frac{1}{A} \left(1 + \frac{R_f}{R_i} \right) \right)}$$

Electronics

Inverting configuration:



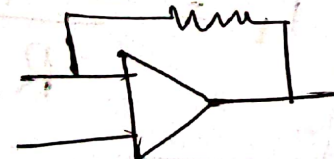
∴ If an Op-Amp does not draw any current,

$$i_{in} = i_f$$

$$\therefore V_1 = V_2 = 0 \text{ (input gain)} \quad \text{--- (i)}$$

$$V_a = V_1, \quad V_b = V_2$$

□ If just  is given, gain A is the open loop gain.

□ But if  $\Rightarrow$ there is a feedback, and gain is closed loop gain. (-ve feedback).

$$\text{Now, } i_{in} = \frac{V_{in} - V_1}{R_i} = \frac{V_{in}}{R_i} \quad [\text{as } V_1 = 0, (i)]$$

$$i_f = \frac{V_1 - V_o}{R_f} = -\frac{V_o}{R_f}$$

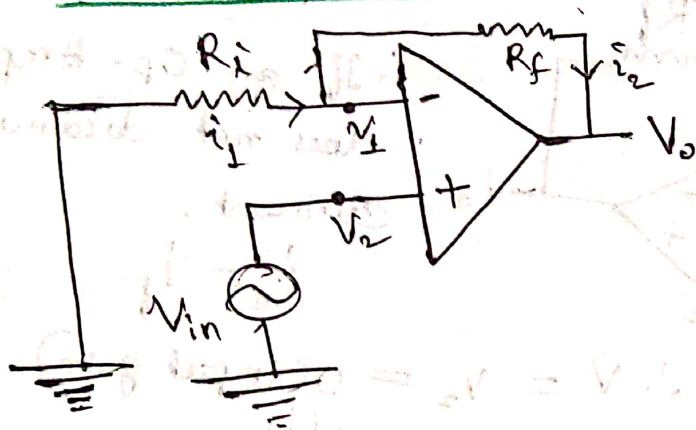
Open loop
 $\rightarrow$ typically ∞ gain

$$\Rightarrow i_{in} = i_f \Rightarrow \frac{V_o}{V_{in}} = -\frac{R_f}{R_i} = A \rightarrow \text{closed loop}$$

If open loop gain is finite with gain:

$$A = -\frac{R_f}{R_i} \left(1 + \frac{1}{A} \left(1 + \frac{R_f}{R_i} \right) \right)$$

▫ Non-Inverting configuration



▫ Case 1: Open loop gain $(A) \rightarrow \infty$
 $V_1 = V_2$ and $i_1 = i_2$

$$i_1 = \frac{0 - V_1}{R_i} \Rightarrow i_1 = -\frac{V_1}{R_i} = i_2$$

▫ How can we control input impedance of say, a CE Transistor?

Ans: By changing Bias, reverse biasing C.B junction and F.B, the E-B junctions. $\rightarrow$ in BJT

In FET, as bias voltage is increased, voltage "pinches off" ; narrowing down the channels of current flow.

$$\text{So, } V_1 - V_o = i_2 R_2 = i_1 R_2 = -\frac{V_1 R_2}{R_1}$$

$$V_o = V_1 \left(1 + \frac{R_2}{R_1} \right)$$

$$\therefore \frac{V_o}{V_1} = \left(1 + \frac{R_2}{R_1} \right) = A \quad (\text{closed loop gain})$$

When open loop gain is finite, closed loop gain :

$$\frac{V_o}{V_1} = \frac{\left(1 + \frac{R_2}{R_1} \right)}{1 + \frac{1}{A} \left(1 + \frac{R_2}{R_1} \right)} \quad (\text{HW: derivation})$$

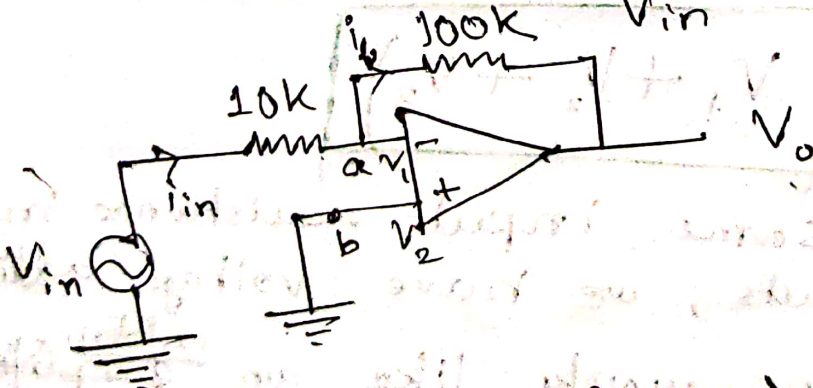
Design of Op-amp having :

(i) $V_o = -10 V_{in} \rightarrow$ infinite gain (inverting config.)

(ii) $V_o = 5 V_{in} \rightarrow$

(i) $V_o = -10 V_{in}$, closed loop gain (A)

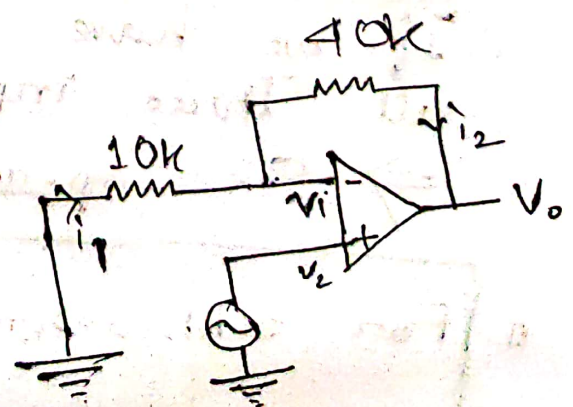
$$= \frac{V_o}{V_{in}} = -10 = -\frac{R_2}{R_1}$$



$$(ii) A = \frac{V_o}{V_1} = 5 = \left(1 + \frac{R_2}{R_1} \right)$$

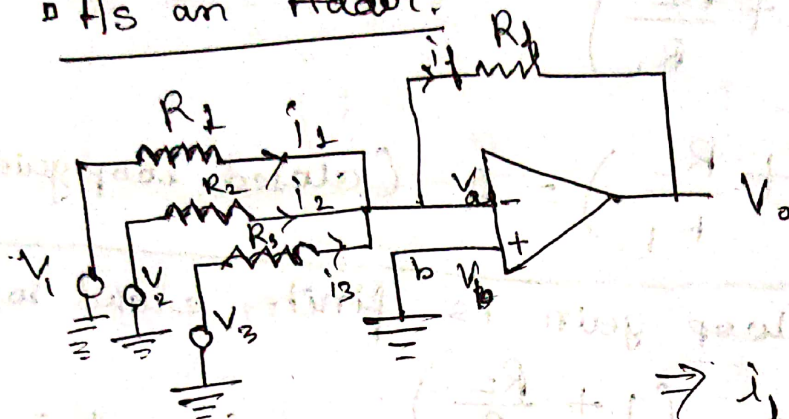
$$R_2 + R_1 = 5 \Omega$$

$$R_2 = 4 R_1$$



▢ Roles of Op-Amp:

▢ As an Adder:



$$\therefore i_{in} = i_1 + i_2 + i_3$$

$$V_a - V_o = i_f R_f = 0$$

$$\Rightarrow i_f = -\frac{V_o}{R_f} = i_1 + i_2 + i_3$$

$$\Rightarrow i_1 = \frac{V_1 - V_a}{R_1} + \frac{V_2 - V_a}{R_2} + \frac{V_3 - V_a}{R_3}$$

$$\Rightarrow -\frac{V_o}{R_f} = \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3}$$

$$\Rightarrow V_o = -R_f \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right)$$

Suppose we put $R_1 = R_2 = R_3 = R$

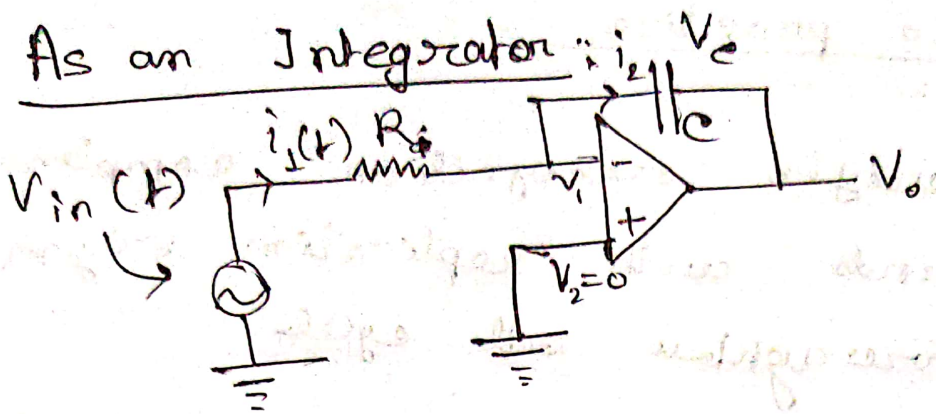
$$\Rightarrow V_o = -\frac{R_f}{R} (V_1 + V_2 + V_3)$$

∴ If we have same input resistance in all three inputs, we have voltage addition

⇒ So op-amp works like an adder.

▢ For subtractor → 1 input inverting, other in non-inverting.

As an Integrator: i_2 V_c



► If we apply Ohm's law:

$$V_{in} - V_1 = i_1(t) R$$

$$V_o = -V_c = -\frac{q}{C} = -\frac{1}{C} \int i_2 dt$$

$$= -\frac{1}{C} \int i_1(t) dt \quad [\because i_1(t) =$$

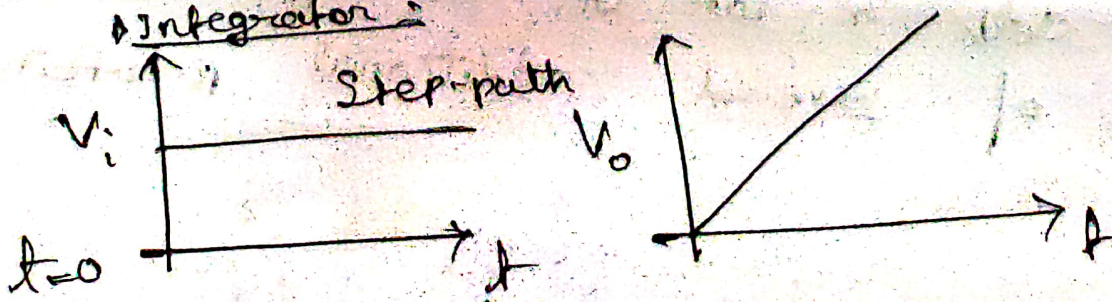
$$V_o = -V_c = -\frac{1}{C} \int \frac{V_{in}(t)}{R} dt$$

$$\Rightarrow V_o = -\frac{1}{RC} \int V_{in}(t) dt$$

For differentiator, capacitor and resistor positions are exchanged in integrator configuration → Homework: Formulate V_o equation for differentiator.

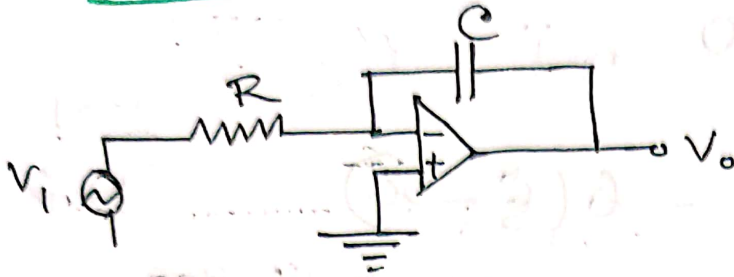
Integrator / Differentiator !

Integrator:



□ Op-Amp;

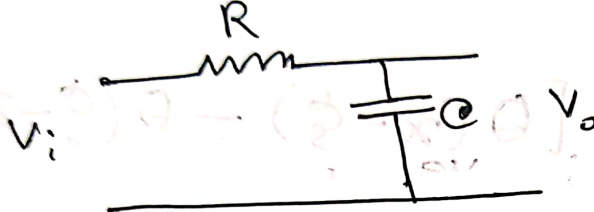
▷ Integration Circuit



For op-amp.

$$V_o = -\frac{1}{RC} \int_0^t V_i(t) dt$$

□ Note: Consider the circuit:



$$V_o = \frac{q}{C} = \frac{1}{C} \int_0^t I dt = \frac{1}{RC} \int_0^t V_i dt$$

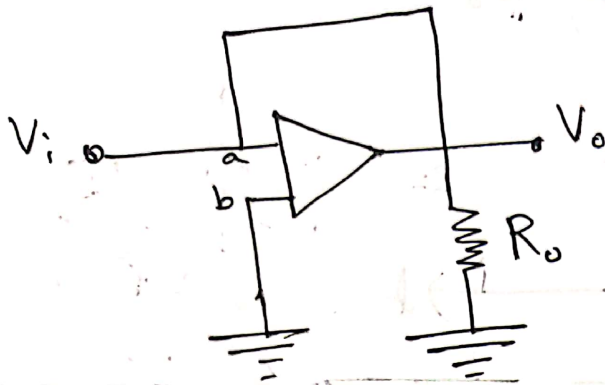
▷ What is the difference between using an R-C circuit and Op-Amp?

□ For Op-Amp, we can use integrator circuit at all times.

at $t=0$, $V_o=0$, $I = \frac{V_i - V_o}{R} = \frac{V_i}{R}$

$$\Rightarrow V_o = \frac{1}{RC} \int_0^t V_i dt \rightarrow \text{not possible for any other time, other than } t=0$$

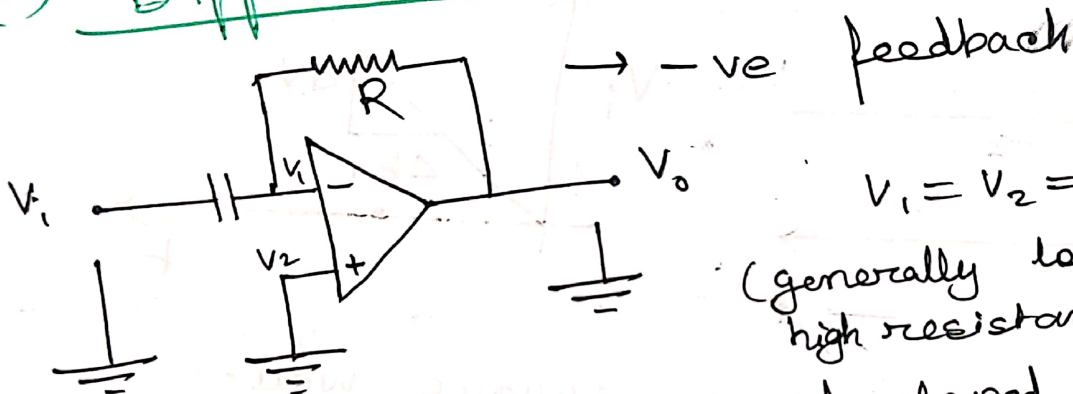
(V) Voltage Follower : (Recall $\rightarrow$ op-Amp has a high input impedance)



Can be used as a Buffer amplifier

$\rightarrow$ To handle impedance mismatch, much like common collector.

(vi) Differentiation



$$V_1 = V_2 = 0$$

(generally low capacitance, high resistance).

Let 'q' be the charge developed in the capacitor due to the input signal.

$$V_i = \frac{q}{C} \Rightarrow \frac{dV_i}{dt} = \frac{1}{C} \frac{dq}{dt} = \frac{I}{C}$$

$$I = \frac{V_i - V_o}{R} = -\frac{V_o}{R}$$

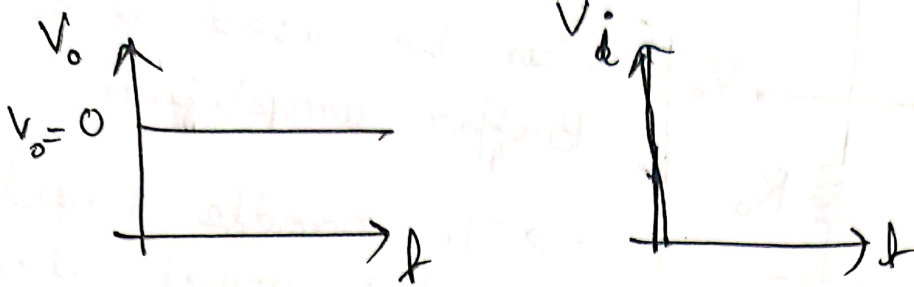
$$= -\frac{V_o/R}{C}$$

$$V_o = -RC \frac{dV_i}{dt}$$

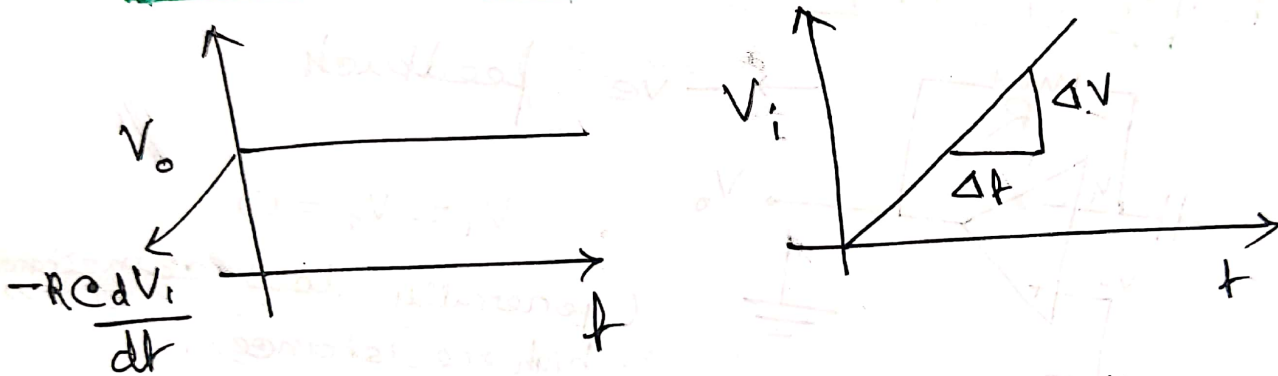
$$V_i = -\frac{V_o}{RC}$$

Examples: (i) If our input V_i is a step function:

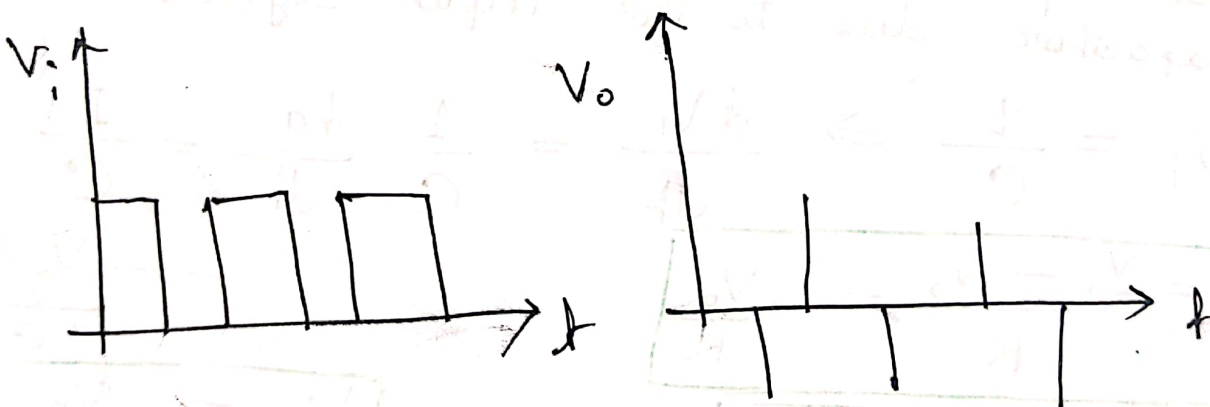
Slope: $\frac{dV_i}{dt} = 0 \Rightarrow V_o = 0$



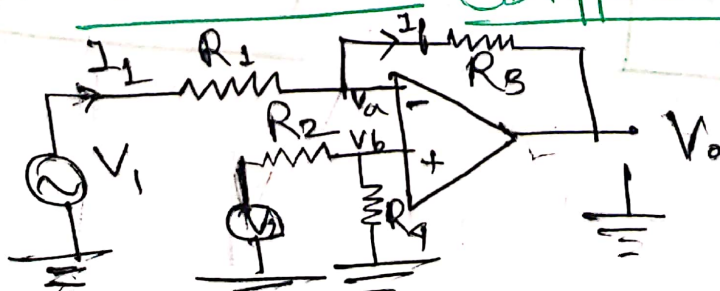
(ii) If our input, V_i is a ramp
Then output voltage is a step function:



(iii) If our input is square wave:



(vii) Subtraction (Difference circuit)



$$I_1 = \frac{V_1 - V_a}{R_1} \quad (i)$$

$$I_2 = \frac{V_2 - V_b}{R_2} \quad (ii)$$

$$I_d = (V_a - V_o)/R_3 \quad (iii)$$

also, $\frac{V_b - V_G}{R_4} = I_2$ or $I_2 = \frac{V_b}{R_4}$ — (iv)
 At the op-amp summing point:

$V_a = V_b$
 From (iii) and (iv)

$$\frac{V_b}{R_4} = I_2 = \frac{V_2 - V_b}{R_2}$$

$$\Rightarrow V_b \left(\frac{1}{R_b} + \frac{1}{R_2} \right) = \frac{V_2}{R_2}$$

$$\Rightarrow V_b = \left(\frac{R_4}{R_2 + R_4} \right) V_2 \quad \text{--- (v)}$$

If $V_2 = 0$, then $V_b = V_a = 0 \Rightarrow V_{o(a)} = V_1 \left(-\frac{R_3}{R_1} \right)$

If $V_1 = 0$, $V_b = V_a = V_2 \left(\frac{R_4}{R_2 + R_4} \right)$ (using (v)) — (vi)

• From op-Amp $I_1 = I_f \Rightarrow \frac{V_1 - V_a}{R_1} = \frac{V_a - V_o}{R_3}$
 (from eqn (i)) (from (vi))

• Now $V_{out(b)}$:

(Recall $V_a = V_2 \left(\frac{R_4}{R_2 + R_4} \right)$)

$$\frac{-V_a}{R_1} = \frac{V_a - V_{out}}{R_3}$$

$$\frac{V_{out(b)}}{R_3} = V_a \left(\frac{R_1 + R_3}{R_3 R_1} \right) \Rightarrow V_a \left(\frac{R_1 + R_3}{R_1} \right) = V_{out(b)}$$

$$\Rightarrow V_{out(b)} = \left(\frac{R_1 + R_3}{R_1} \right) \left(\frac{R_4}{R_2 + R_4} \right) V_2$$

$$V_{out} = V_{out(a)} + V_{out(b)}$$

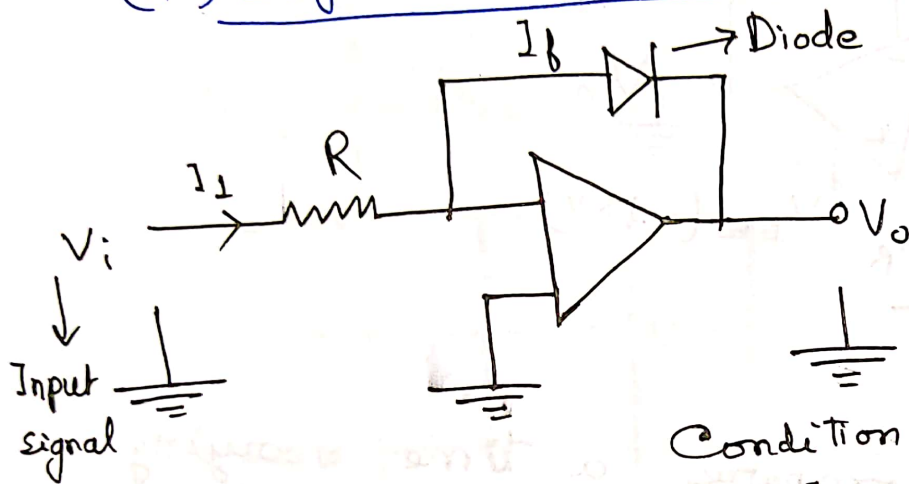
$$V_o = -\frac{R_3}{R_1} V_1 + V_2 \left(\frac{R_1 + R_3}{R_1} \right) \left(\frac{R_4}{R_2 + R_4} \right)$$

when $R_1 = R_2$, $R_3 = R_4$

$$V_{out} = \left(\frac{R_3}{R_2} \right) (V_2 - V_1)$$

□ OP-Amp:

(viii) Logarithmic Operator:



Condition for op-amp requires $I_1 = I_f$

□ Diode current:

$$I_f = I_0 (e^{eV/kT} - 1)$$

□ For large V_f , $e \gg k_B T$

$$\frac{k_B T}{e} = V_T \Rightarrow I_f = I_0 e^{V_f/V_T}$$

$$\Rightarrow \frac{V_f}{V_T} = \ln \left(\frac{I_f}{I_0} \right) = \ln(I_f) - \ln(I_0)$$

$$I_f = I_1 = \frac{V_{in}}{R} \quad \text{and} \quad V_f = -V_o$$

$$\Rightarrow -V_o = V_T \left[\ln \left(\frac{V_{in}}{R} \right) - \ln I_0 \right]$$

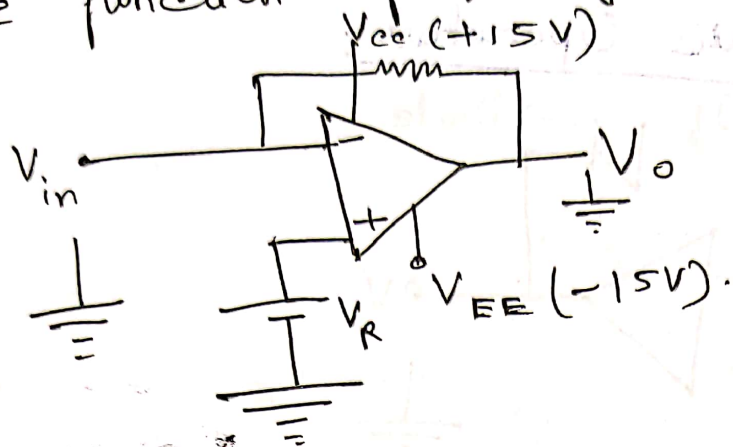
$$\Rightarrow V_o = -V_T \left[\ln \left(\frac{V_{in}}{R} \right) - \ln I_0 \right]$$

$$\Rightarrow \boxed{V_o = -\frac{k_B T}{e} \ln \left(\frac{V_{in}}{I_0 R} \right)}$$

→ Output is proportional to the logarithm of input voltage.

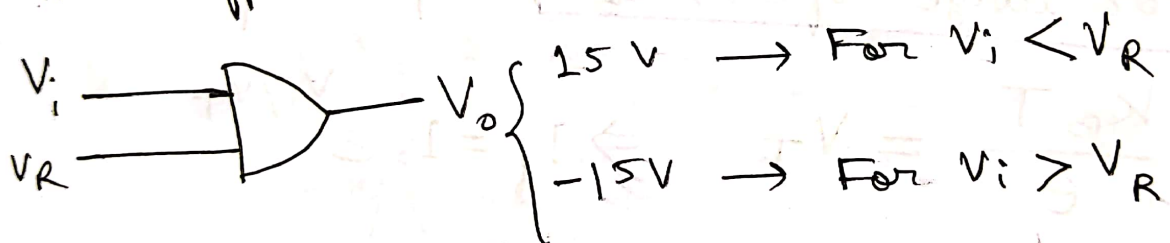
(i x) Voltage Comparator

The function of voltage comparator is:



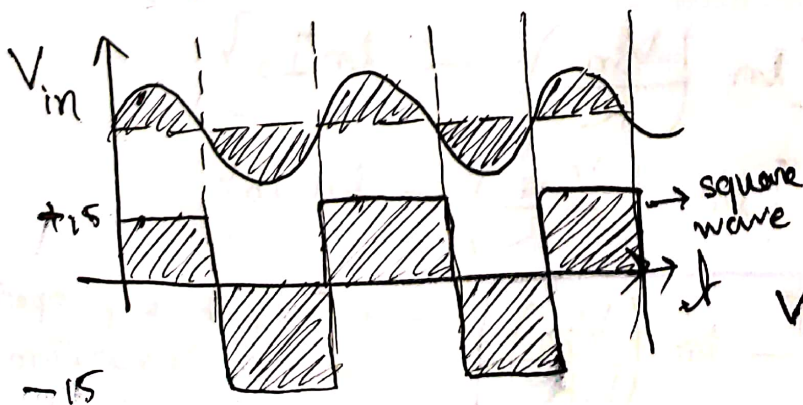
► Function: To compare a time varying voltage at one input with a fixed reference voltage at the other (non-inverting) output. Saturation voltage is 15 V here.

o The difference that amplifier input/output:



$$V_o = A(V_R - V_{in}) = +15V \Rightarrow [V_{in} < V_R]$$

$$V_o = -15V \quad [V_R < V_{in}]$$



If A is the internal gain of op-amp.

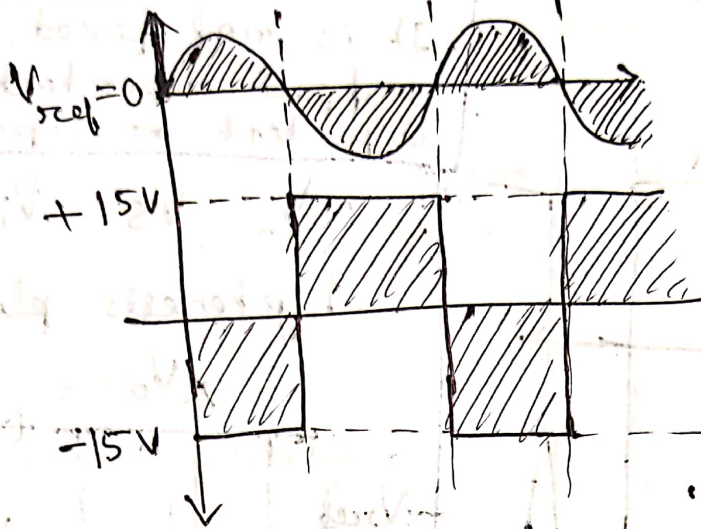
$$V_D = V_R - V_{in}$$

→ Differential input.

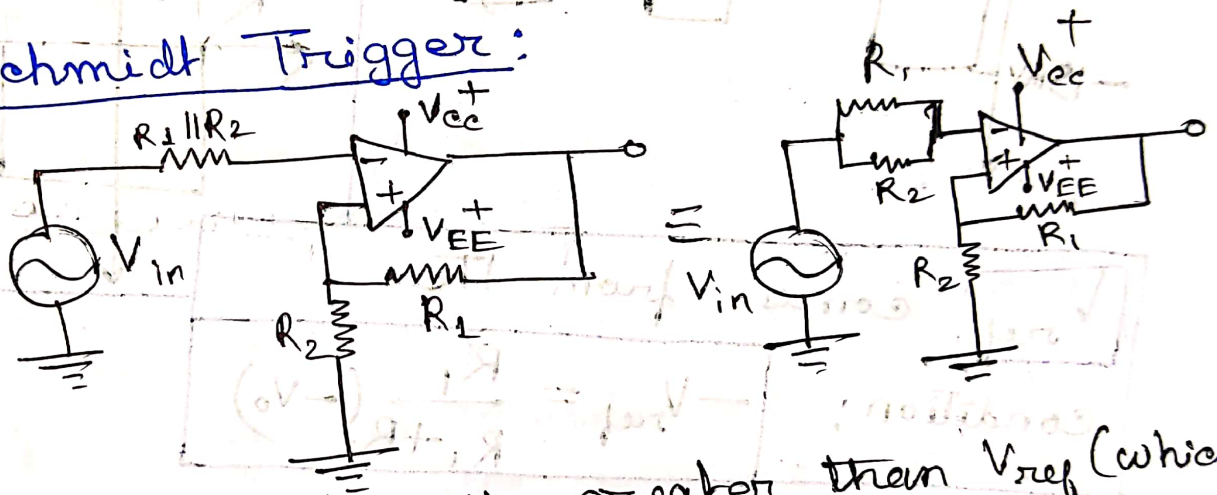
$$V_o = AV_D = A(V_R - V_i)$$

□ Case I: When $V_i > V_R$, Then $V_o = -ve$, The output switches to $-V_{sat} = -15V$

□ Case II: When $V_i < V_R$: $V_o = (V_R - V_i)A$
 $\rightarrow +ve \rightarrow +15V$



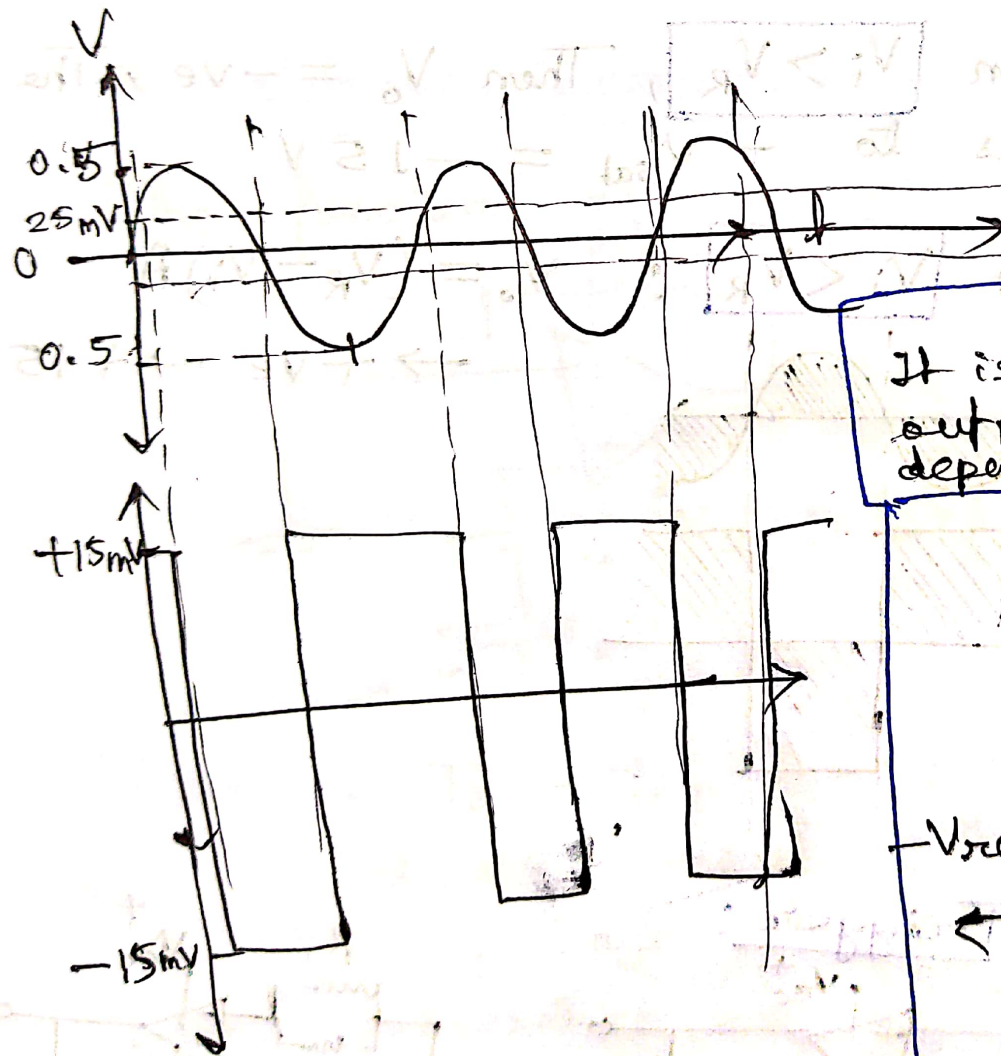
(X) Schmidt Trigger:



• When V_{in} is slightly greater than V_{ref} (which is not zero in this case), the output becomes $-V_{sat}$ and if V_{in} is slightly less than $-V_{ref}$ (more negative than $-V_{ref}$), then the output becomes V_{sat} .

$$V_b = \frac{R_1}{R_2 + R_1} V_2 \Rightarrow V_{ref} = \frac{R_1}{R_1 + R_2} V_o$$

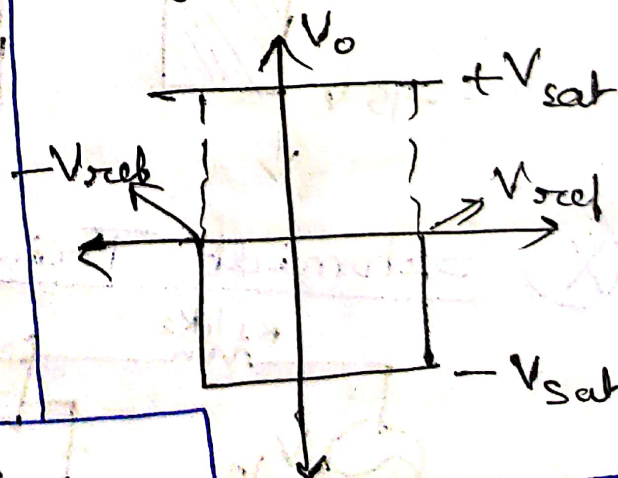
$$-V_{ref} = \frac{R_1}{R_1 + R_2} (-V_o) \quad [P.T.O.]$$



25mV is the V_{ref} here.

or threshold.
It is not fixed, but output dependent, not dependent on input voltage

V_o V_S V_{in}
hysteresis plot



V_{ref} comes from the condition:
$$-V_{ref} = \frac{R_1}{R_1 + R_2} (-V_o)$$

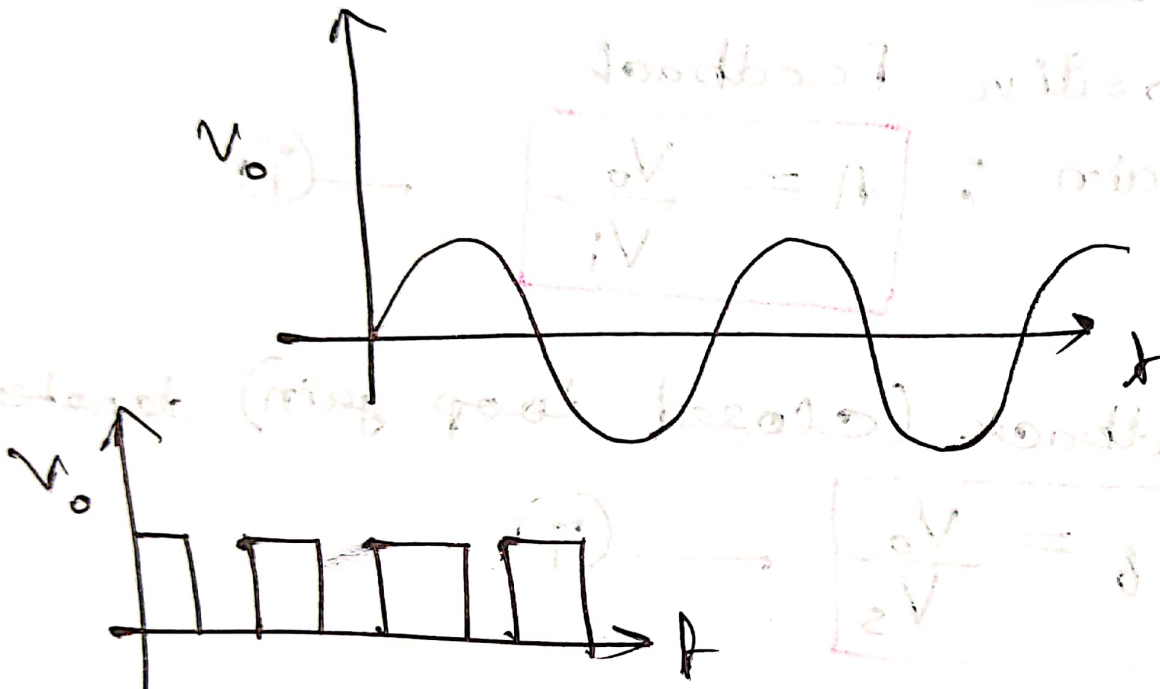
Electronics :

Lecture 39

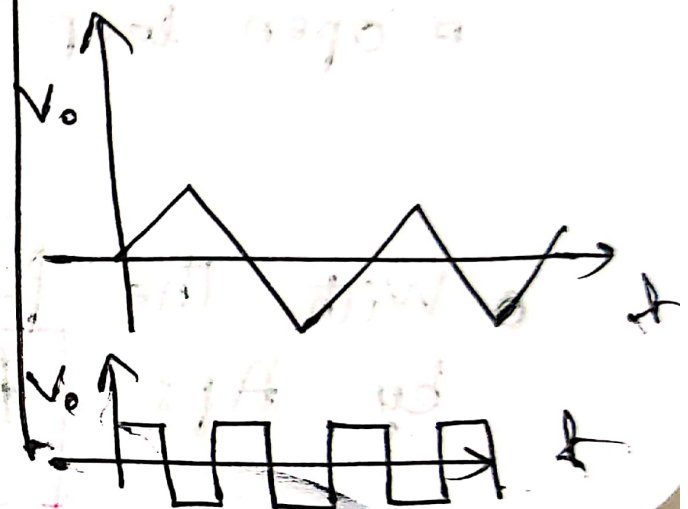
□ Oscillator :

- An oscillator is a circuit which is used to generate an A.C. voltage without any input signal ($V_{in} = V_{sig} = 0$)

o We want a sinusoidal output :



□ Other kinds of outputs :

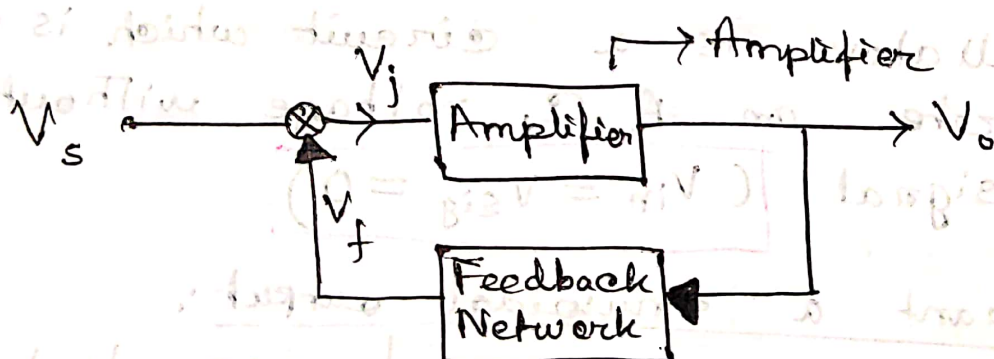


▪ Sinusoidal Oscillation : Can be categorised.

- (a) Audio frequency oscillator (upto 20 kHz).
- (b) Radio frequency (r.f.) (20 kHz to 30 MHz).
- (c) VHF (very high frequency) (30 MHz to 300 MHz).
- (d) UHF (Ultra high frequency) (300 MHz to 3 GHz).
- (e) Microwave frequency oscillator (Above 3 GHz).

▷ Block Diagram :

▪ Amplifier : CE Amplifier, FET Amplifier, OP Amplifier.



Positive Feedback

▪ open loop gain : $A = \frac{V_o}{V_i}$ — (i)

• With the feedback (closed loop gain) denoted by A_f : $A_f = \frac{V_o}{V_s}$ — (ii)

• The feedback is positive and the voltage V_f is added to V_s to generate V_i .

$$V_i = V_s + V_f \quad \text{--- (iii)}$$

Defining the feedback gain: --- (iv)

$$\beta = \frac{V_f}{V_o} ; \text{ From eqn (iii), } V_i = V_s + \beta V_o$$

$$\Rightarrow V_i = V_s + \beta (A V_s) \quad \text{(from eqn (i))}$$

$$\Rightarrow V_s = V_i (1 - A\beta)$$

$$\text{But, } A_f = \frac{V_o}{V_s} = \frac{V_o}{V_i (1 - A\beta)} = \frac{A}{(1 - A\beta)}$$

$$\text{Here, } |A_f| > |A|$$

Here $|A_f| > A \Rightarrow$ The product of the open loop gain and the feedback factor is called loop gain (AB)

$$\text{If } AB = 1, \quad A_f = \infty = \frac{V_o}{V_s}$$

$$\Rightarrow V_s = 0$$

The gain of the amplifier with the +ve feedback is infinite, and the amplifier gives an A.C. output without any input signal. The condition $\Rightarrow AB = 1 \rightarrow$ Barkhausen condition

Phase shift will be 360° .

Types of Oscillator:

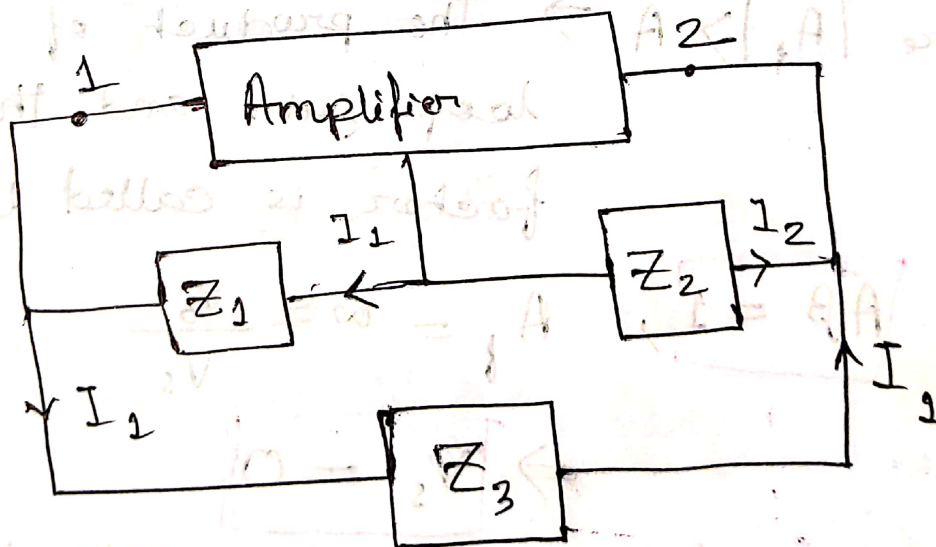
- (i) RC oscillator
- (ii) LC oscillator
- (iii) Crystal oscillator

AF
RF
VHF

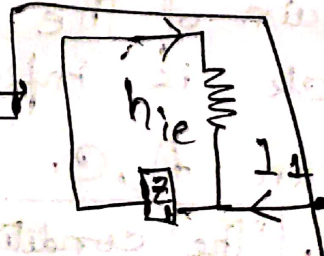
Caution about frequency stability:

- (i) Temperature fluctuations.
- (ii) Power fluctuations, $\rightarrow$ Good Voltage regulation takes care of this.
- (iii) A-factor

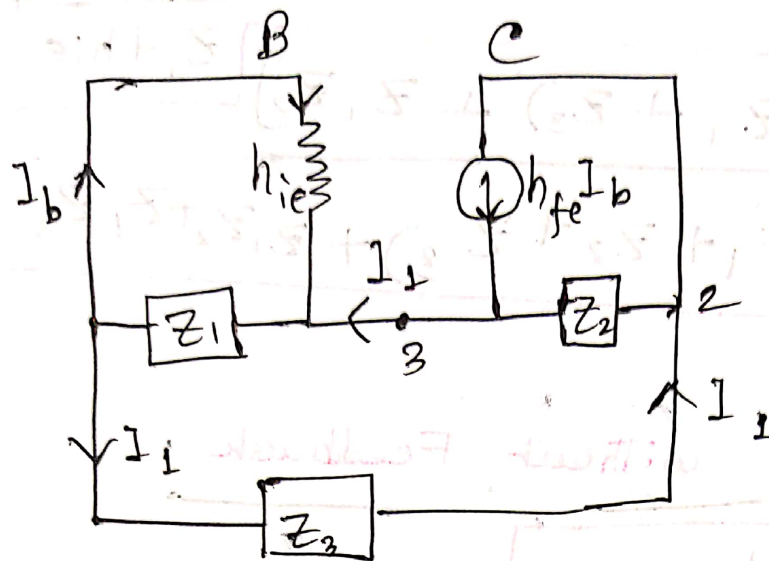
L.C. Oscillator:



Hybrid equivalent circuit



Hybrid equivalent circuit:



- Z_1 and Z_2 act as AC voltage divider for the output voltage and the feedback signal.
- Input terminals are 1 and 3, output terminals are 2 and 3.

Load impedance:

Since Z_1 and the input resistance h_{ie} of the transistor are in parallel, their equivalent impedance Z' is

$$Z' = Z_1 \parallel h_{ie}$$

$$\frac{1}{Z'} = \frac{1}{Z_1} + \frac{1}{h_{ie}} \Rightarrow Z' = \frac{Z_1 h_{ie}}{Z_1 + h_{ie}} \quad \text{--- (1)}$$

The load impedance Z_2 between the output terminals '2' and '3' is the equivalent impedance of $Z_2 \parallel Z'$ and Z_3 . $Z_i = Z_2 \parallel (Z' + Z_3)$

$$\frac{1}{Z_L} = \frac{1}{Z_2} + \frac{1}{(Z' + Z_3)} = \frac{1}{Z_2} + \frac{1}{\frac{Z_1 h_{ie}}{Z_1 + h_{ie}} + Z_3}$$

$$\Rightarrow Z_L = \frac{Z_2 [h_{ie}(Z_1 + Z_3) + Z_1 Z_3]}{h_{ie}(Z_1 + Z_2 + Z_3) + Z_1 Z_2 + Z_1 Z_3} \quad (ii)$$

o Voltage gain without Feedback :

$$A_v = \frac{-h_{fe} Z_L}{h_{ie}} \quad (iii)$$

o Feedback fraction β .

The output voltage between 2 and 3 is given in terms of I_1 ,

$$V_o = -I_1 (Z' + Z_3) = -I_1 \left[\frac{Z_1 h_{ie}}{Z_1 + h_{ie}} + Z_3 \right]$$

$$\Rightarrow V_o = -I_1 \left[\frac{Z_1 h_{ie}}{Z_1 + h_{ie}} + Z_3 \right]$$

$$\Rightarrow V_o = -I_1 \left\{ \frac{h_{ie}(Z_1 + Z_3) + Z_1 Z_3}{Z_1 + h_{ie}} \right\}$$

Voltage feedback to the input terminals '1' and '3' is given by:

$$V_o = -I_1 Z' = -I_1 \left[\frac{Z_1 h_{ie}}{Z_1 + h_{ie}} \right] \quad \text{--- (v)}$$

$$\beta = \frac{V_o}{V_i} = I_1 \left[\frac{Z_1 h_{ie}}{Z_1 + h_{ie}} \right] \times \left[\frac{Z_1 + h_{ie}}{h_{ie}(Z_1 + Z_3) + Z_1 Z_3} \right] \frac{1}{I_1}$$

$$\beta = \frac{Z_1 h_{ie}}{h_{ie}(Z_1 + Z_3) + Z_1 Z_3} \quad \text{--- (vi)}$$

$$V_o = -I_1 (Z' + Z_3)$$

$$\Rightarrow V_o = -I_1 \left(\frac{Z_1 h_{ie}}{Z_1 + h_{ie}} + Z_3 \right)$$

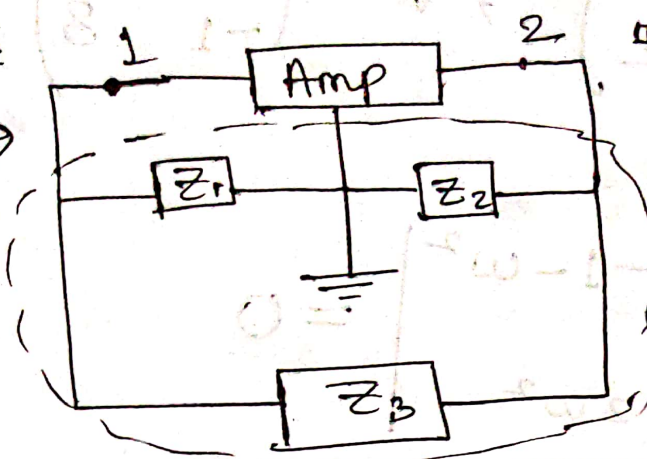
□ Barkhausen Criteria:

$$h_{ie}(Z_1 + Z_2 + Z_3) + Z_1 Z_2 (1 + h_{fe}) + Z_1 Z_3 = 0$$

Electronics Mathematical Methods for Physics

Lecture
no-40

Ckt 1



For L-C oscillator

We drew the equivalent circuit last class.

For C.E.

$$I_c = \beta I_b$$

For CE

180° phase shift between V_o , V_{in} (For CE)

$V_o \rightarrow$ is phase shifted from input by

180°.

$$A = - \frac{V_o}{V_{in}}$$

So, Barkhausen Criteria:

1st $\rightarrow A\beta = 1$

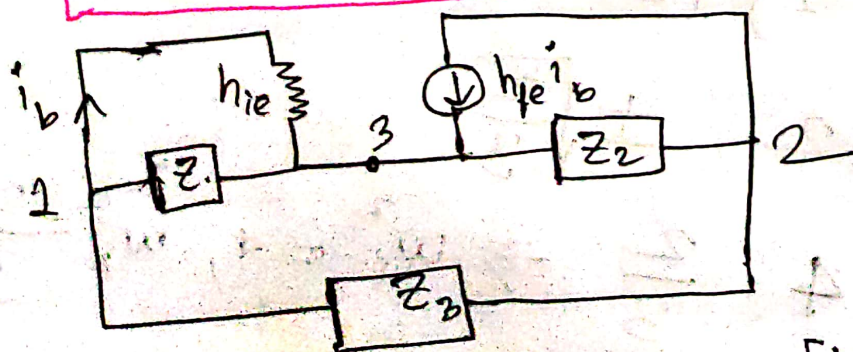
$$A_v = \frac{A}{1 - A\beta}$$

2nd $\rightarrow$ Total phase shift between V_o and V_{in} should be

360°

(depending on type of transistor)

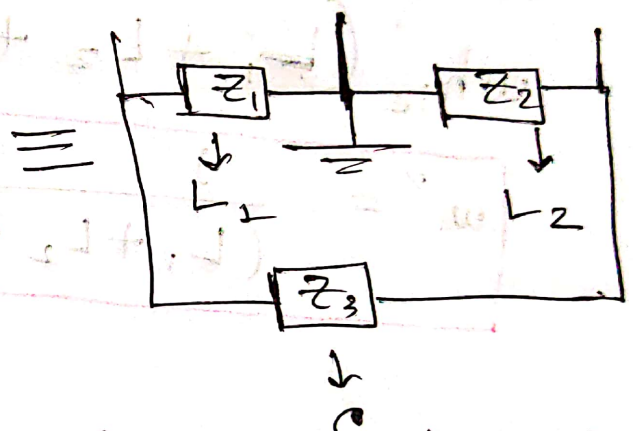
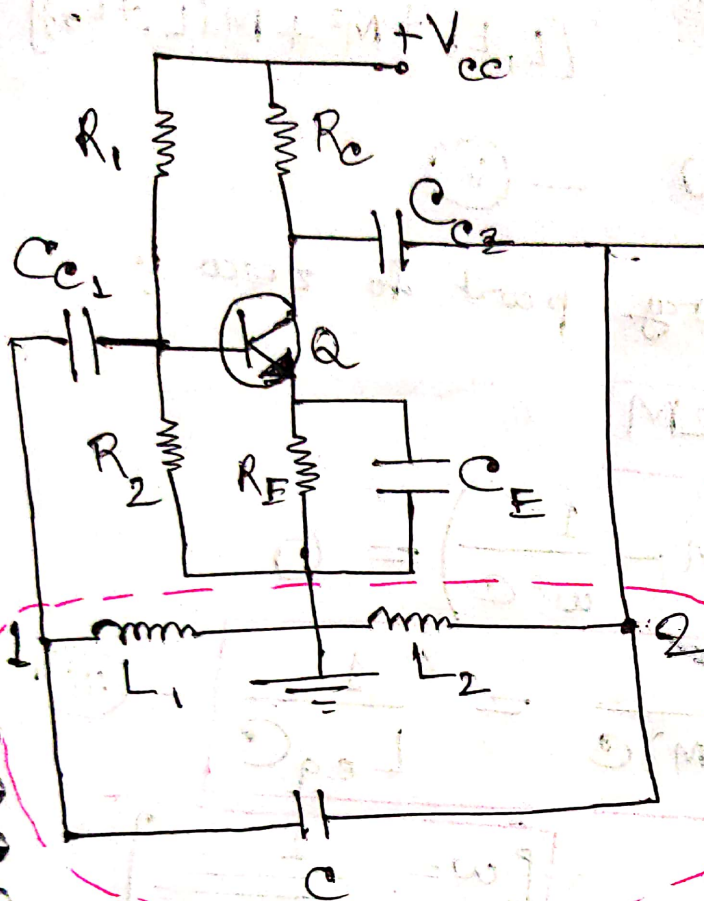
$$h_{ie}(Z_1 + Z_2 + Z_3) + Z_1 Z_2 (1 + h_{fe}) + Z_1 Z_3 = 0 \quad (i)$$



The feedback has to be designed in a way that it gives a 360° phase shift

[What happens when no inversion?]

▫ Hartley Oscillator :



▫ There is a mutual inductance between L_1 and L_2 (Equivalent to ckt 1)

▸ eqn of the L.C. oscillator

▫ $Z_1 = j\omega L_1 + j\omega M$ [$M \equiv$ Mutual inductance]

▫ $Z_2 = j\omega L_2 + j\omega M$

▫ $Z_3 = \frac{1}{j\omega C} = -\frac{j}{\omega C}$

Substituting in equation (i) :

$$\Rightarrow h_{ie}(j\omega L_1 + j\omega M + j\omega L_2 + j\omega M - \frac{j}{\omega C})$$

$$+ (1 + h_{ie})(j\omega L_1 + j\omega M)(j\omega L_2 + j\omega M)(j\omega L_1 + j\omega M)\left(\frac{-j}{\omega C}\right) = 0$$

$$\Rightarrow j\omega h_{fe} \left[L_1 + L_2 + 2M - \frac{1}{\omega^2 C} \right] + (1 + h_{fe}) j^2 \omega^2 [L_1 L_2 + M^2 + M(L_1 + L_2)]$$

$$- j^2 \left[\frac{L_1}{C} + \frac{M}{C} \right] = 0 \quad \text{--- (ii)}$$

equating the imaginary part to zero :

$$L_{eq} = L_1 + L_2 + 2M$$

$$\text{while } \left(L_1 + L_2 + 2M - \frac{1}{\omega^2 C} \right) = 0$$

$$\omega^2 = \frac{1}{(L_1 + L_2 + 2M)C} = \frac{1}{L_{eq}C} \quad \text{--- (iii)}$$

$$\left[\omega = \frac{1}{\sqrt{L_{eq}C}} \right]$$

Substituting (iii) in (ii),

$$j\omega h_{fe} (L_1 + L_2 + 2M - L_1 - L_2 - 2M)$$

$$+ (1 + h_{fe}) j^2 \omega^2 [L_1 L_2 + M(L_1 + L_2) + M^2]$$

$$+ \frac{L_1}{C} + \frac{M}{C} = 0$$

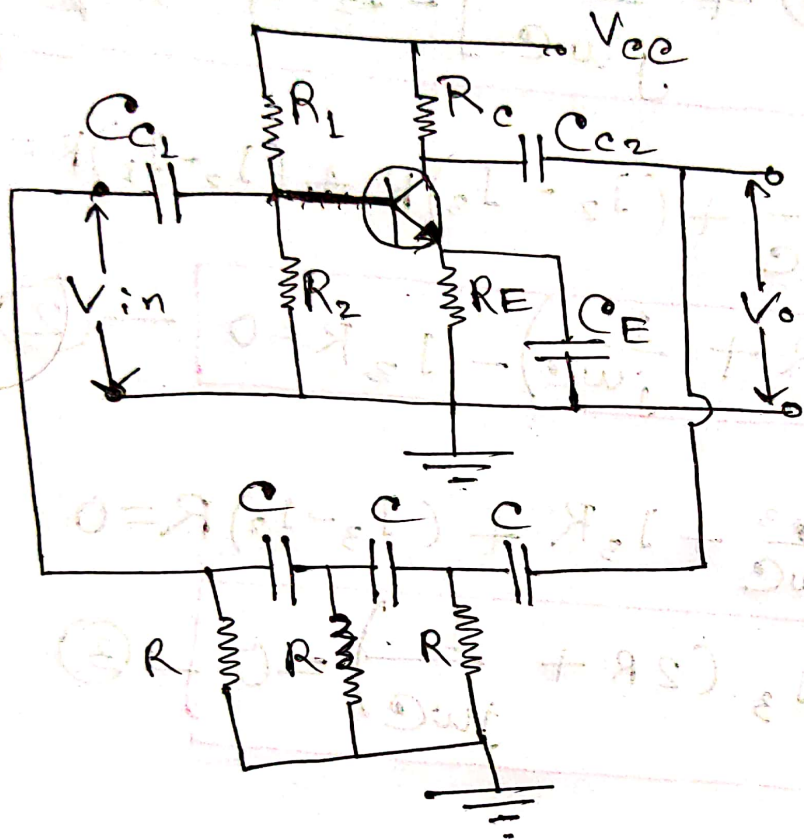
$$\therefore h_{fe} = \frac{L_1 + 2L_1 M + M^2}{L_1 L_2 + M(L_1 + L_2) + M^2}$$

$$\Rightarrow h_{fe} = \frac{(L_1 + M)^2}{(L_1 + M)(L_2 + M)} = \frac{L_1 + M}{L_2 + M}$$



So we need $h_{ie} \geq \frac{L_1 + M}{L_2 + M}$ for the Barkhausen Criteria to be fulfilled.

RC phase shift oscillator:

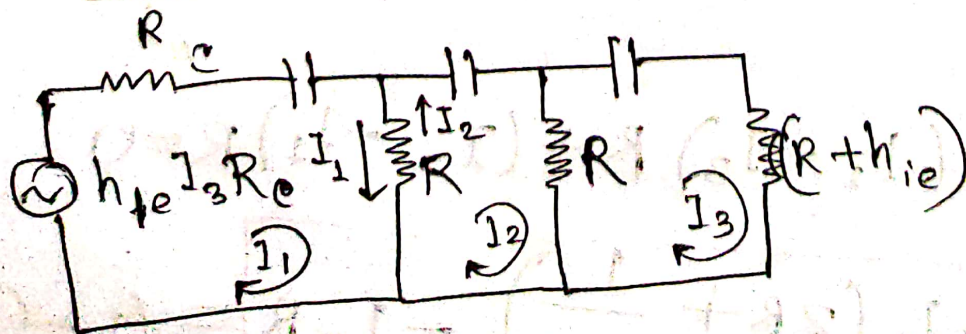
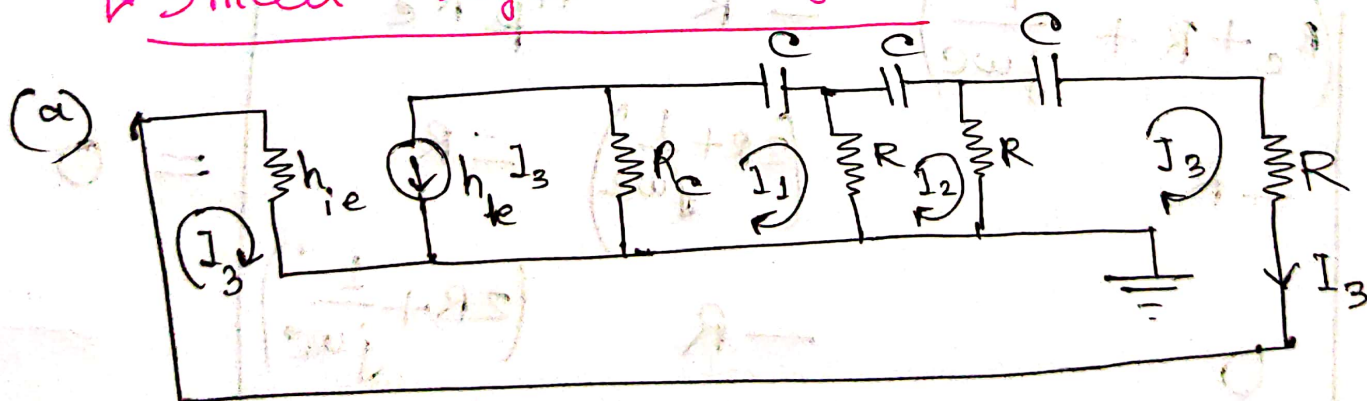


Phase shift:

$$\Phi = \tan^{-1} \left(\frac{1}{\omega CR} \right)$$

each R-C network should give us a phase of 60° .

Small Signal Analysis:



We have $I_N = h_{fe} I_3$, $R_N = R_e$ in $\text{ckt}(a)$.

In loop 1, $-h_{fe} I_3 R_e = I_1 R_e + \frac{I_1}{j\omega C} + (I_1 - I_2)R$

$$I_1 \left[(R_e + R) + \frac{1}{j\omega C} \right] - I_2 R + I_3 (h_{fe} R_e) = 0 \quad \text{--- (1)}$$

In loop 2, $\frac{I_2}{j\omega C} + (I_2 - I_3)R + (I_2 - I_1)R = 0$

$$\Rightarrow -I_1 R + I_2 \left(2R + \frac{1}{j\omega C} \right) - I_3 R = 0 \quad \text{--- (2)}$$

In loop 3, $\frac{I_3}{j\omega C} + I_3 R + (I_3 - I_2)R = 0$

$$\Rightarrow -I_2 R + I_3 \left(2R + \frac{1}{j\omega C} \right) = 0 \quad \text{--- (3)}$$

$$\begin{vmatrix} \left(R_e + R + \frac{1}{j\omega C} \right) & -R & h_{fe} R_e \\ -R & \left(2R + \frac{1}{j\omega C} \right) & -R \\ 0 & -R & \left(2R + \frac{1}{j\omega C} \right) \end{vmatrix} = 0$$

$$\Rightarrow \frac{I_2}{j\omega C} + (I_2 - I_3)R + (I_2 - I_1)R = 0$$

$$\Rightarrow -I_1 R + I_2 \left(2R + \frac{1}{j\omega C} \right) - I_3 R = 0 \quad \text{--- (2)}$$

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RC Phase Shift Oscillation:

We solve the determinant and get:

$$(R + R_c + \frac{1}{j\omega C}) \left(2R + \frac{1}{j\omega C} \right) \left(2R + \frac{1}{j\omega C} \right) - R^2$$

$$-(-R) \left\{ -R \left(2R + \frac{1}{j\omega C} \right)^2 - 0 \right\} + h_{fe} R_c R^2 = 0$$

$$\Rightarrow \left[R + R_c + \frac{1}{j\omega C} \right] \left\{ 4R^2 + \frac{4R}{j\omega C} + \frac{1}{j^2 \omega^2 C^2} - R^2 \right\}$$

$$- R^2 \left(2R + \frac{1}{j\omega C} \right) + R^2 h_{fe} R_c = 0$$

Real part = 0

$$(R + R_c) \left\{ 3R^2 - \frac{1}{\omega^2 C^2} \right\} - \frac{4R}{\omega^2 C^2} - 2R^3 + R^2 h_{fe} R_c = 0$$

Imaginary part to 0,

$$-(R + R_c) \frac{4R}{\omega C} - \frac{1}{\omega C} \left[3R^2 - \frac{1}{\omega^2 C^2} \right] + \frac{R^2}{\omega C} = 0$$

$$\Rightarrow \omega^2 = \frac{1}{C^2 [6R^2 + 4RR_c]}$$

$$\Rightarrow f = \frac{1}{2\pi RC} \frac{1}{\sqrt{6 + 4k}}$$

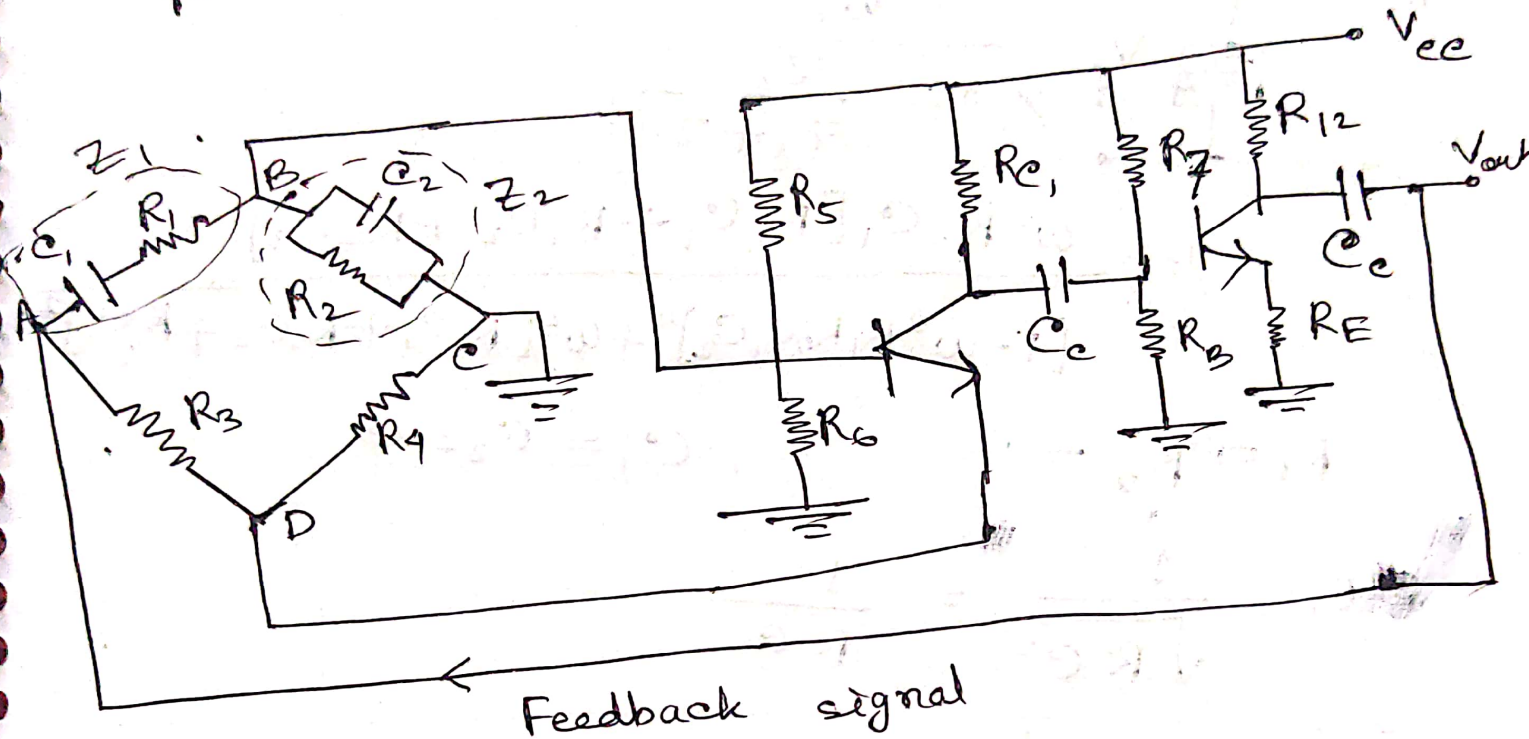
Gain of the amplifier: $\frac{29R}{R_c} + 23 + \frac{4R_c}{R}$

$$h_{fe} = \frac{29}{k} + 23 + 9k$$

[P.T.O.]

$$\frac{d h_{fe}}{d k} = 0 \rightarrow k = \sqrt{\frac{29}{4}} = 2.6925$$

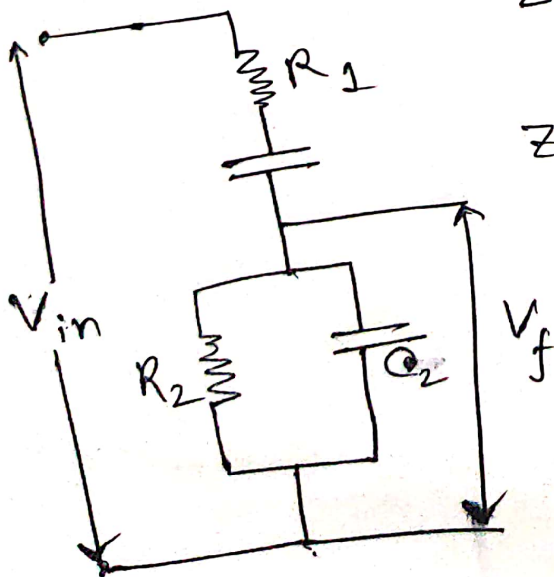
$h_{fe} \text{ (min)} = 44.53 \cdot h_{fe} \geq 44.55 \rightarrow$ Barkhausen criterion.



Lead-lag network.
 For low frequency, the phase leads. For high frequency, the phase lags.

$$Z_1 = R_1 + \frac{1}{j\omega C_1} = \frac{1 + j\omega R_1 C_1}{j\omega C_1}$$

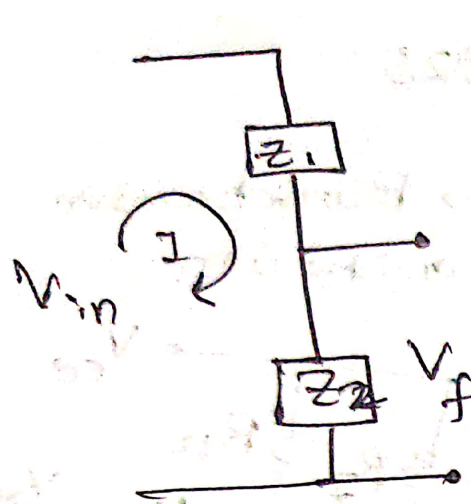
$$Z_2 = R_2 \parallel \frac{1}{j\omega C_2} = \frac{R_2}{1 + j\omega R_2 C_2} \quad \text{--- (ii)}$$



$$\beta = \frac{V_f}{V_{in}}$$

$$\omega = \frac{1}{\sqrt{R_1 R_2 C_1 C_2}}$$

How?



$$I = \frac{V_{in}}{Z_1 + Z_2}$$

$$V_f = I Z_2$$

$$\Rightarrow V_f = \frac{V_{in} Z_2}{Z_1 + Z_2}$$

$$\beta = \frac{V_f}{V_{in}}$$

$$\Rightarrow \beta = \frac{Z_2}{Z_1 + Z_2}$$

$$\beta = \frac{\omega^2 R_2 C_1 [R_1 C_1 + R_2 C_2 + R_3 C_3]}{[1 - \omega^2 R_1 R_2 C_1 C_2]^2 + \omega^2 [R_1 C_1 + R_2 C_2 + R_3 C_3]^2}$$

$$R_1 = R_2 = R, \quad C_1 = C_2 = C$$

$$\omega = \frac{1}{\sqrt{R^2 C^2}} = \frac{1}{RC}$$

$$\beta = \frac{j \frac{1}{RC} R_2 C_1}{(1-1) + j \left(\frac{1}{RC} R_1 C_1 + \frac{1}{RC} R_2 C_2 + \frac{1}{RC} R_2 C_1 \right)}$$